

Everest Engineering College

Sanepa, Lalitpur

Assignment 4 & 5 , 2024

Level : BE First Semester

Subject: Calculus I

Due Date: Feb 2, 2025

Students are required to give their answers in their own words as far as practicable and copy from others is strictly Prohibited.

Attempt all the Questions:

1. Show that

(a) $\int_0^{\pi/2} \frac{\sqrt{\cot x}}{1+\sqrt{\cot x}} dx = \pi/4$

(b) $\int_0^{\pi/2} \frac{\sin x - \cos x}{1 + \sin x \cos x} dx = 0$

(c) $\int_0^{\pi/2} \log(\sin x) dx = \int_0^{\pi/2} \log(\cos x) dx = \frac{\pi}{2} \log(\frac{1}{2})$.

(d) $\int_0^{\pi/4} \log(1 + \tan \theta) d\theta = \frac{\pi}{8} \log 2$

(e) $\int_0^1 \frac{\log(1+x)}{1+x^2} dx = \frac{\pi}{8} \log 2$

(f) $\int_0^1 \cot^{-1}(1-x+x^2) dx = \frac{\pi}{2} - \log 2$

2. Evaluate the integral, if exists.

(a) $\int_0^\infty \frac{1}{(1+x^2)^{3/2}} dx$

(b) $\int_0^\infty e^{-ax} \sin bx dx$

(c) $\int_0^1 \frac{\sin^{-1} x}{\sqrt{1-x^2}} dx$

(d) $\int_0^\infty \frac{\log x}{x^2} dx$

(e) $\int_0^a \sqrt{\frac{a-x}{x}} dx$

(f) $\int_{-\infty}^\infty \frac{1}{x^3} dx$

3. Show that the integral $\int_2^\infty \frac{\cos^2 x}{x^2} dx$ is convergent by using comparison test.

4. Show that the integral $\int_3^\infty \frac{1}{1+e^x} dx$ is convergent but $\int_3^\infty \frac{1}{1-e^{-x}} dx$ is divergent by using comparison test.

5. Show that the integral $\int_2^\infty \frac{x^2}{\sqrt{x^7+1}} dx$ is convergent but $\int_2^\infty \frac{x^3}{\sqrt{x^7-1}} dx$ is divergent by using comparison test.

6. Obtain the reduction formula for $\int \cos^n x dx$ and hence evaluate $\int \cos^7 x dx$ and $\int_0^{\pi/2} \cos^7 x dx$

7. Find the reduction formula for $\int \cot^n x dx$ and hence evaluate $\int \cot^7 x dx$.

8. Using gamma function, show that
 - i. $\int_0^1 x^2(1-x^2)^{3/2} dx = \pi/32$.
 - ii. $\int_0^a \frac{x^4}{\sqrt{a^2-x^2}} dx = \frac{3a^4\pi}{16}$.
 - iii. $\int_0^1 \frac{1}{(1-x^6)^{1/6}} dx = \pi/3$.
 - iv. $\int_0^{2a} x^{9/2} \sqrt{2a-x} dx = \frac{21a^6\pi}{16}$.
9. Show that the perimeter of a circle $x^2 + y^2 = a^2$ is $2\pi a$.
10. Find the arc length of $y^2 = 4ax$ to the points cut off by line $3y = 8x$.
11. Show that the length of the curve $y = \log(\frac{e^x-1}{e^x+1})$ from $x = 1$ to $x = 2$ is $\log(e + \frac{1}{e})$.
12. Show that the length of the arc of parabola $y^2 = 4ax$ from the vertex to one extremity of the latus rectum is $a(\sqrt{2} + \log(1 + \sqrt{2}))$.
13. Find the length of the curve $y^2 = (2x-1)^3$ cut off by the line $x = 4$.
14. Find the area Bounded by $y = x^2$ between $x = 1$ and 3 .
15. Find the area bounded by the curve $y = 2 - x^2$ and $y = -x$.
16. The area bounded by the curve $y^2 - 4x = 4$ and the line $4x - y = 16$
17. Find the area bounded by $x = y^3$ and $x = y^2$
18. Find the area bounded by $y = \sin \frac{\pi x}{2}$ and $y = x$.
19. Find the area bounded by the curves $x - y^3 = 0$ and $x - y = 0$
20. Find the area bounded by the curves $y = x^2 + 1$ and line $y = -x + 3$
21. Find the area bounded by y-axis and $x = y^2 - y^3$
22. Find the area bounded by $y = \sin x$, $y = \cos x$ and y-axis in the first quadrant.
23. Find the area bounded by the curve $y = x^3 - 12$ and x-axis.
24. show that the area of the astroid $x^{2/3} + y^{2/3} = a^{2/3}$ is $\frac{3}{8}\pi a^2$.
25. Find the area between the curves $x^2 = 4ay$ and $y^2 = 4ax$.
26. Find the area of the region and the circle $x^2 + y^2 = 4$ cut off by the lines $x - 2y = -2$ in the first two quadrants. **[Hint: example 14]**.
27. Find the area bounded by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$
28. Find the area of the surface obtained by rotating the curve about x-axis
 - i. $y = \sqrt{x}$ $4 \leq x \leq 9$.
 - ii. $y = \frac{x^3}{6} + \frac{1}{2x}$ $1/2 \leq x \leq 1$.
29. Find the area of the sphere generated by revolution of the circle $x^2 + y^2 = r^2$ about x-axis.
30. Find the area of the surface generated by rotating the portion of the curve $y = \frac{1}{3}(x^2 + 2)^{3/2}$ between $x = 0$ and $x = 3$ about y-axis.
31. The loop of the curve $9x^2 = y(3-y)^2$ is rotating about x-axis. Find the surface area generated.

32. Find the area of the surface obtained by rotating the curve $y = \cos x$, $0 \leq x \leq \frac{\pi}{3}$ about x-axis.
33. Show that the area of the curved surface generated when one loop of the curve $x^2(a^2 - x^2) = 8a^2y^2$ is revolved about x-axis is $\frac{\pi a^2}{4}$.
34. Find the volume of solids generated by revolving the region bounded by the lines and the curves $y = \sqrt{\cos x}$, $0 \leq x \leq \pi/2$, $y = 0$, $x = 0$ about x-axis.
- 35.
36. Find the volume of solids generated by revolving the region bounded by the lines and the curves (i) $y = \sqrt{2 \sin 2x}$, $0 \leq y \leq \pi/2$, $x = 0$ (ii) $x = y^{3/2}$, $x = 0$, $y = 2$ about y-axis.
37. The circle $x^2 + y^2 = a^2$ revolves around the x-axis, show that the volume of the whole sphere generated is $\frac{4\pi a^3}{3}$.
38. Prove that the volume of the ellipsoid formed by revolution of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ about the x-axis is $\frac{4\pi ab^2}{3}$.
39. Find the volume of the solid generated by revolving the region bounded between $x^2 = 4y$ and $y = |x|$ about $y = -2$.
40. Find the volume of the solid generated by revolution of the region bounded by $y = \sqrt{x}$ and the lines $y = 1$, $x = 4$ about the line $y = 1$. [Hint: example 24].
41. Find the volume of the solid generated by revolving the region bounded by $y = \sqrt{x}$ and the lines $y = 2$, $x = 0$ about the line $y = 2$.
42. Find the volume of the solid generated by revolving the region bounded on the left by the line $x = 4$ and on the right by the circle $x^2 + y^2 = 25$ about y-axis.
43. Find the volume of the solid in the region in the first quadrant bounded by the curve $y = x^2$ below by x-axis and on the right by the line $x = 1$ about the line $x = -1$.

"ALL THE BEST."